

# RealQM on Conduction

Claes Johnson<sup>1</sup>

<sup>1</sup>KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden ,  
claesjohnson@gmail.com

## Abstract

RealQM, as an alternative to textbook quantum mechanics, represents an  $N$ -electron system by  $N$  real-valued wave functions  $\psi_i(x)$ , with  $x$  a coordinate in three-dimensional Euclidean space, whose squares  $\psi_i^2$  are unit charge densities with non-overlapping supports meeting at free interfaces. The physics is carried by a single energy functional: electron Coulomb potential energy without self-repulsion, plus kinetic energy. No other force enters it and nothing in it is fitted. The ground state is the configuration minimising that functional over the supports as well as over the wave functions, so the interfaces are unknowns of the problem rather than inputs to it, and the cost is polynomial of small power in  $N$ .

RealQM has already been applied to solids — ionic crystals under melting and under shear, molecular solids including ice and solid nitrogen, a lithium lattice of a hundred atoms recovering from a perturbation under its own dynamics — and, further afield, to molecular binding, hydrogen bonds and solvation, to ionisation energies across the periodic table, and to nuclear binding by dual Coulomb confinement. All of it is *bound* matter, where an attractor is present and the charge is where the attractor is, and in none of it was transport attempted.

The task posed here is conduction, the first question asked of RealQM about charge that no attractor holds in place: the cost of advancing the electron pattern of a solid by one lattice site, which is the activation energy for charge transport. The answer is obtained from Coulomb interaction and kinetic energy alone, with nothing fitted, and no such computation appears to have been carried out before. The reason is structural rather than a want of effort. In textbook quantum mechanics the  $N$ -electron state is a function on a  $3N$ -dimensional configuration space, an object beyond computation for any  $N$  of interest, so every route that can actually be followed first reduces the dimension and then repairs the reduction with modelled parameters — a relaxation time, a transfer integral, an effective mass — and the transport energy comes out of those parameters rather than out of the two terms it is made of. Working in three dimensions from the start removes the need for them. That is the new capability reported here: the quantity is computed rather than modelled.

Computationally the difficulty is one of scale. Atomic energies are tens of electron-volts, semiconductor activation is 50–500 meV per carrier and metallic conduction sets in near 20 meV, while the total energy of a chain long enough to pose the question is thousands of electron-volts. The barrier sought is therefore between  $10^{-3}$  and  $10^{-5}$  of the number a calculation returns, depending on the arrangement. It is not reached by computing that number to five figures, but by differencing two configurations of the same system — the same electrons, the same kernels, the same grid, differing only in where a few interior domain walls sit — so that every large contribution to the error is common to both and cancels.

Three arrangements of charge are computed here and a fourth, a cubic lattice, in §8; the barrier to advancing the electron pattern by one site differs by nearly three orders of magnitude across the three. These are not one-dimensional models. The wave functions are functions on ordinary three-dimensional space, the domains are regions of it, and the calculations run on grids of some two to three million points; what is arranged along a line is the nuclei. A **chain of**

**hydrogen atoms**, one electron to each kernel, gives  $\approx 394$  meV per electron and a polarisability  $\alpha = 7.6N$ , extensive in the number of electrons. Its landscape is not the assumed one: scanned across a period, the minimum lies not at registry but near a quarter spacing, 196 meV below the arrangement with each electron centred on its own kernel, and the two maxima are of unequal height, so advancing one site is a two-step passage rather than a single crossing. A **cable of Li ions**, the carriers standing off closed cores, gives  $\approx 246$  meV. Those two are the computed values, at the coefficient  $\frac{1}{2}$  that the kinetic term carries throughout. Varying that coefficient is then a sensitivity and not a result: raised to 0.75 the cable's barrier is 66 meV, at 0.9 it is 20 meV — below room temperature's  $kT = 26$  meV, so a cable so stiffened would not hold its charge — and at 1.0 it is 1 meV. No value was sought that reproduces anything measured, and the third arrangement is reported to show how soft the barrier is in this one coefficient, not to exhibit a conductor. The preferred registry, carriers on their own ion sites, is unchanged across the range, so what the stiffening weakens is the pinning itself and not merely its location.

What is computed is an energetics of charge, and one transport quantity follows from it with nothing added. A uniform field tilts the landscape and nothing more, so the field at which the barrier disappears is a property of  $E(d)$  alone:  $5.2 \times 10^9$ ,  $3.2 \times 10^9$  and  $2.6 \times 10^8$  V m $^{-1}$  for the three arrangements. The first two are above the dielectric strength of silica and would break down before they would slide; the stiffened cable is below it and would slide first.

A conductivity requires more, and the division is the ordinary one: an energy landscape from a theory of charge, a rate from a theory of statistics laid over it, as density functional theory is routinely used with transition-state theory. In the activated regime two things must be supplied, a clock and a temperature, and with them the drift velocity follows from the landscape directly. Only the temperature is an import the framework could not in principle replace, a frequency being available from nuclei that already move classically. Taken on those terms the barriers give room-temperature conductivities of 0.1 and 10 S m $^{-1}$ , which bracket intrinsic germanium at 2 S m $^{-1}$ . The attempt frequencies come out at  $h\nu = 76$  and 23 meV, lattice energies that were not asked for and a check that the mass entering them is of the right kind; but they are computed from the curvature a sinusoid would have, and the scan shows the real well to be much flatter than that, so both frequencies — and the conductivities with them — are overestimates by a factor this article does not fix. The barriers are upper bounds for a reason independent of mechanism — a barrier is a minimum over paths joining two registries and the rigid slide is one such path — so the conductivities are lower bounds by a factor this article does not determine.

## 1 The basic question posed to RealQM

RealQM represents an  $N$ -electron system in terms of  $N$  real-valued wave functions  $\psi_i(x)$ , with  $x$  a coordinate in three-dimensional Euclidean space, whose squares  $\psi_i^2$  are unit charge densities with non-overlapping supports  $\Omega_i$  meeting at free interfaces. The physics of the system is described through its energy functional, the sum of electron Coulomb potential energy without self-repulsion and kinetic energy, with no other force entering and nothing fitted. Explicitly,

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \left( \frac{1}{2} |\nabla \psi_i|^2 - \sum_s \frac{Z_s}{|x - x_s|} \psi_i^2 \right) dx + \frac{1}{2} \sum_{i \neq j} \iint \frac{\psi_i^2(x) \psi_j^2(y)}{|x - y|} dx dy, \quad (1)$$

where  $i$  and  $j$  index the electrons,  $s$  indexes the nuclei,  $Z_s$  and  $x_s$  are the charge and position of nucleus  $s$ , and  $y$  is a second space coordinate; the supports are disjoint and  $\int_{\Omega_i} \psi_i^2 = 1$ . Atomic units are used throughout: lengths in Bohr radii  $a_0 = 0.529$  Å, energies in hartree, 1 Ha = 27.211 eV.

The ground state of the system is the configuration minimising this energy, the minimisation running over the supports as well as over the wave functions, so that the interfaces are unknowns of the problem and not inputs to it; excited states are the other stationary points of the same energy. The cost of minimising (1) is polynomial of small power in  $N$ .

Every previous test of RealQM — ionisation energies across the periodic table [22], molecular binding, hydrogen bonds and solvation [19, 21], nuclear binding by dual Coulomb confinement [23] — has been on *bound* matter, where an attractor is always present and the charge is where the attractor is. Conduction is the first question that asks about charge which is not where an attractor holds it, and the framework has no separate machinery for it: the same energy, the same domains, in a different arrangement of the same two kinds of charge.

The task posed to it here is accordingly this: what does it cost to advance the electron pattern of a solid by one lattice site, which is the activation energy for charge transport. What enters the answer is the two terms of (1) and a specification of the nuclei, with nothing adjusted to reproduce a measured quantity, and such a computation does not appear to have been carried out before, for reasons §2 sets out.

What is computed is the energetics, and the limit of that is best put dimensionally: a minimisation of (1) yields energies, while a current is charge per time and a conductivity is charge per time per field. No quantity of energy supplies a time. So a rate theory must be brought in from outside, here as everywhere else, and §7 is exact about which parts of that import the framework might one day supply for itself — the time scale being the part that a lattice already moving classically could provide.

**Three-dimensional throughout.** Both arrangements below place their nuclei on a line, and it is worth saying at once that nothing else about them is one-dimensional. Each  $\psi_i$  is a function on  $\mathbb{R}^3$ , each support  $\Omega_i$  is a region of it with a two-dimensional free surface, and the domains tile a three-dimensional box —  $138^3$  and  $148^3$  points for the two systems here. A linear chain of nuclei is the smallest arrangement in which a registry between a carrier lattice and an ion lattice can be posed at all, which is why it is used; it is not a reduction of the electronic problem, and there is no step anywhere in this article at which a dimension is integrated out.

**Two arrangements, both ordinary matter.** The first is a **chain of atoms**: one kernel per site with its valence electron occupying the same region, which is what a hydrogen chain is and what an alkali is to a first approximation. The second is a **cable of ions surrounded by electrons**: kernels screened by their own core electrons, with the valence charge standing off outside them, which is how a wire is built. The question is what each gives.

## 2 What textbook quantum mechanics offers

Recall that textbook quantum mechanics is based on Schrödinger’s equation in  $3N$  dimensions for a single antisymmetric wave function  $\Psi(x_1, \dots, x_N)$ , with the exclusion principle carried by the antisymmetry and with a Hamiltonian built, like (1), from Coulomb interaction and kinetic energy alone. But the cost of solving it is exponential in  $N$ , and the models actually used — Hartree, Kohn–Sham and their descendants — always include drastic reductions of the dimensionality. There is thus no textbook treatment of conduction and insulation based on Schrödinger’s equation. There is a stack of models, each carrying quantities supplied from measurement.

Metal is separated from insulator by a gap at the Fermi level [1], and in practice the gap comes from a chosen exchange–correlation functional [2]; the Kohn–Sham gap is not the excitation gap even for the exact functional [3], and the discrepancy is repaired by hybrid mixing or a *GW* self-energy [4]. Transport is then a second model on top of the first: a Boltzmann equation closed by a relaxation time, with an effective mass and a deformation potential for the semiconductor [5]; a transfer integral and a reorganisation energy for a hopping conductor [6, 7]; a localisation

length and a density of states at the Fermi level for variable-range hopping in a disordered one [8]; a Hubbard  $U$  where the band picture fails outright and the material insulates by correlation instead [9, 10]. The recent direction of travel is worth noting, because it runs the other way: learned exchange–correlation functionals [11] replace a handful of constructed parameters with a neural network trained on reference data, and interatomic potentials fitted to density-functional output [12] dispense with the functional altogether in favour of a model fitted end to end. These are effective, and the neural wavefunction methods [13] are a genuine advance on the variational problem — but they solve the  $3N$ -dimensional problem with a better ansatz, and the density-functional line answers the cost of that problem by adding fitted content rather than removing it.

The point of the contrast is not accuracy — these are accurate, and are what one should use to predict a number for a given material. It is that they are several models with parameters, addressing the regimes separately, whereas what follows is one energy with none, in which they differ by how the charge is arranged.

**How the comparison should be read.** Two misreadings are worth heading off, because they pull in opposite directions. The first is to score RealQM against the accuracy of methods that carry fitted quantities. That is the wrong axis: the claim here concerns what the model contains, not whether it beats a percentage, and a framework with no adjustable content should not be judged by one that has some.

The second is to require agreement with a near-exact solution of Schrödinger’s equation — configuration interaction, coupled cluster, quantum Monte Carlo — for the same system. RealQM is not an approximation to that equation. It is a different formulation, and it is not empirically equivalent to it: the two make different predictions, and where they differ the question is which the world agrees with, not which reproduces the other. Disagreement with a benchmark solution of the  $3N$ -dimensional problem is therefore not a defect in a formulation that does not claim to solve it.

What remains, and it is the only thing that adjudicates, is measurement. RealQM’s record there is on bound matter: the binding of  $H_2$  comes out 4.20 eV against 4.75 measured, and ionisation energies across the periodic table are reproduced without fitting [22, 19].

For transport there is as yet nothing, and it is worth being exact about why. The systems computed below are model systems: a chain of eight hydrogen atoms and a cable of sixteen lithium ions, chosen because they are the smallest arrangements in which the question can be posed at all. Neither has an experimental counterpart — nobody has measured the conduction barrier of a linear  $H_8$  chain — so the absence of a comparison is a consequence of what was computed, not a shortcoming of the framework that computed it. Model systems without counterparts are ordinary; the Ising model and jellium are studied for what they isolate, not for what they reproduce.

The consequence is to be accepted rather than dressed. Conductivities of real materials appear below only to set a scale, and no agreement with them is claimed or implied. What would supply a comparison is the activation energy, which is exactly what an Arrhenius plot of a real activated conductor measures, computed for a system that has one. That is a matter of choosing a larger and more realistic cell, which the cost of the method permits; it is not a matter of the method improving.

Equation (1) has no  $3N$ -dimensional object in it and no antisymmetry: the role played by the exclusion principle is played instead by the requirement that the supports do not overlap.

### 3 What RealQM computes

A domain carries one unit of charge by constraint, so no charge passes between sites. What can move is the *partition*: the boundaries are redrawn, and the pattern of domains advances through the lattice. When the pattern has advanced by one full spacing, every electron has taken over its neighbour's kernel and the configuration is restored — identical, since the electrons are identical and the energy depends on the partition and not on labels. Charge has been transported and nothing material has traversed the sample, which is the relation a dislocation bears to slip.

The quantity that decides whether this happens is the energy along that path. Write  $E(d)$  for the energy with the partition displaced by a fraction  $d$  of a spacing. It is periodic,  $E(d+1) = E(d)$ , and its amplitude

$$V_0 = \max_d E(d) - \min_d E(d) \quad (2)$$

is the corrugation: the barrier the partition must cross to advance one site. A field tilts this curve by  $-NeFd$ , so the pattern runs when the tilt exceeds the steepest slope of  $E(d)$ , and otherwise settles at a displacement and stops — bounded polarisation, no current.

Three energies are easily confused here and are worth separating at the outset. The barrier is *invested and returned*: the pattern climbs to  $d = \frac{1}{2}$  and descends again, ending one site advanced, and the net cost of that advance is zero — which is not an argument but the control  $E(1) = E(0)$ . The barrier therefore governs how *often* an advance happens, not how much energy a conductor loses. What a conductor loses is the work done by the field,  $eFa$  per site per advance, which passes into the lattice; it is present whether the barrier is large or small, and a good conductor dissipates more than a poor one because more charge moves. Only the first of the three is computed in this article. The second is verified. The third belongs to the driven regime and needs the dissipation channel §6 says this framework does not supply.

$E(d)$  is *scanned* across a period and (2) read off it. That is worth stating plainly, because it is not what an earlier version of this work did: two values of  $d$  were evaluated, 0 and  $\frac{1}{2}$ , and their difference called the amplitude. Symmetry does place stationary points there — the curve is periodic and inversion-symmetric about a site, so  $E(\frac{1}{2} + x) = E(\frac{1}{2} - x)$  — but it says neither that these are the only ones nor that they are the extremes, and for the chain of §4 they are not: the minimum lies near  $d = \frac{1}{4}$  and the two maxima are of unequal height, so the two-point difference was taken between a maximum and a maximum. The amplitude is the one quantity that matters exponentially, and it cannot be known until the landscape has been seen.

Scanning also removes an assumption from the two quantities read off the shape. The curvature  $K$  that sets the attempt frequency is taken at the true minimum, and the steepest slope that sets the threshold field is the steepest measured slope, rather than both being inferred from a sinusoid fitted to the amplitude. A sinusoid ties them together,  $K = 2\pi^2 V_0/a^2$ ; a real landscape does not, and the chain's steepest slope is some 35% above the sinusoidal value at the same amplitude.

**Two activation criteria, and what they are not.** A corrugation delivers two statements and they are independent. The *amplitude* against  $kT$  decides whether charge is thermally held. The *steepest slope* gives the field at which the tilt  $-eFx$  overcomes the pinning with no thermal help,  $F_{\text{th}} = \max |dE/dx|/e$ , which carries no time and no statistics and so is the one transport quantity a minimisation can supply outright. At fixed amplitude the two move separately: a landscape with the same height but a steeper rise is equally held against heat and harder to free with a field. Neither is a conductivity. The first gives an exponent and not a rate; the second gives a threshold and not a current; and both describe the rigid slide, so both are bounds — the activation energy from above and the conductivity from below.

**The partition is imposed, not found.** Everything below evaluates  $E(d)$  with the domains *given*: the supports are specified as a cell-ownership map and held fixed, and only the wave functions relax within them. In the cable that means the boundary between one carrier and the next, and the boundary between a core and its carrier, are both prescribed. The states computed are therefore constrained minima of (1), not ground states of it, and §1’s statement that the interfaces are unknowns of the problem describes the framework rather than these calculations.

Two reasons, of different kinds. The first is structural:  $E(d)$  is a function of a displacement, so  $d$  must be specified for the question to mean anything — the partition is the coordinate being scanned, exactly as a constrained coordinate maps a reaction barrier, and nobody supposes the reacting system is constrained. This is why translation by one full spacing is the control: two specified configurations related by an exact symmetry of the lattice must return the same energy, and that test has no force unless the partition is specified. The second reason is a limitation rather than a choice, and §8 returns to it: a released boundary sits in one cell or the next, so a sub-threshold force moves it by zero cells or by one, and what a released calculation reports is that quantisation rather than the physics.

**Ends, and how to remove them.** A finite chain has ends, and they dominate: terminal domains bind an electron-volt harder than interior ones, so a corrugation read from total energies is reading the ends. Displacing only the middle  $W$  carriers repairs this. The energy difference is then  $W$  times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly, and no terminal domain moves in either configuration:

$$\Delta E(W) = W \varepsilon_{\text{bulk}} + 2\varepsilon_{\text{junc}}, \quad \varepsilon_{\text{bulk}} = \frac{\Delta E(W_2) - \Delta E(W_1)}{W_2 - W_1}. \quad (3)$$

The decomposition asserts linearity in  $W$ , which a third window tests rather than fits.

**Why tens of meV are available out of a hundred hartree.** The two are not the same kind of quantity, and it is worth separating them before the objection is answered. The hartrees are the total energy of the whole system — forty-eight electrons on sixteen sites — whereas the barrier is quoted *per carrier*. What is actually differenced is  $\Delta E(W_2) - \Delta E(W_1)$ , of order  $3 \times 10^{-2}$  Ha, and dividing by  $W_2 - W_1$  turns it into the per-carrier figure. So the quantity to be resolved is about  $2 \times 10^{-4}$  of the total rather than  $10^{-6}$ .

That is still four significant figures down, and the natural objection is that no calculation resolves it. The objection would be right if the total energy had to be accurate. It does not. What is required is that the *difference* be accurate, and the two configurations entering it are the same system: the same domains, the same electrons, the same kernels, the same total charge, on the same grid, differing only in where a few interior walls sit. Every large contribution to the discretisation error — the nuclear regions, the cores, the box — is identical in both and subtracts out. This is the ordinary reason binding energies are quoted in meV while the total energies they come from carry errors a thousand times larger.

That the cancellation actually occurs is not assumed but checked, in four independent ways. The solver is deterministic: repeated runs of one configuration agree to the last reported digit, so there is no stochastic floor beneath the quantity being computed. Translation by one full period is an exact relabelling and must return the same energy, which it does to nine digits — two separately relaxed states agreeing to that precision. The junction cancellation in (3) is exact algebra rather than an approximation, so the instrument introduces no error of its own. And the linearity the decomposition asserts is tested against a third window, which it could fail and does not. What the

last of these does not do is prove the result correct; symmetry controls verify self-consistency, and a systematic error respecting the lattice symmetry would pass all of them. They establish that the arithmetic is sound, which is a smaller claim than the result being right, and a necessary one.

## 4 A chain of atoms

Sixteen hydrogen atoms in a row at spacing  $a = 4.5 a_0$ : kernel charge +1, one electron per site. There is no core, so there is no radius to impose and nothing excluded by a rule — each electron is kept off its neighbours by non-overlap alone. The system is also the standard benchmark of electronic structure: the linear hydrogen chain has been computed across a range of spacings by density-matrix renormalisation, auxiliary-field quantum Monte Carlo and coupled cluster together [17], and its dimerisation and insulator-to-metal transition mapped [18]. At the spacing used here the chain is far on the insulating side of that transition, which is the answer the corrugation below returns. No energy computed here is set against that equation of state, and §A says why: the nuclear cutoff cancels between two configurations of one system, which is what a corrugation is, and not between a chain and a free atom, which is what a binding energy is. This is the simplest arrangement in which the carrier occupies its own kernel’s region, and it is a real system rather than a model atom.

**The landscape, and where its minimum is.** Scanned across the half period, windowed at  $W = 4$  and  $W = 8$  so that the two junctions cancel, the chain gives per electron and relative to registry

| $d$                               | 0 | 0.125 | 0.25 | 0.375 | 0.5  |
|-----------------------------------|---|-------|------|-------|------|
| $\varepsilon_{\text{bulk}}$ (meV) | 0 | −93   | −196 | +36   | +199 |

and by the symmetry  $E(d) = E(1 - d)$  that is the whole period. The amplitude is therefore

$$V_0 = \max_d \varepsilon - \min_d \varepsilon \approx 394 \text{ meV per electron,}$$

and the two features of the curve are worth as much as the number. The minimum is *not* at registry: it lies near  $d = \frac{1}{4}$ , a fifth of an electron-volt below the arrangement in which each electron is centred on its own kernel. And the two maxima are of unequal height, 0 at registry and +199 meV at half-spacing, so advancing one site is a two-step passage over alternating barriers of 394 and 196 meV with a metastable rest point between. No sinusoid represents this, which is why §3 scans rather than assuming.

That the preferred registry is off-centre is a statement about this family of partitions and not yet about the functional. The domains here are slabs displaced rigidly together, so the scan explores one one-parameter family; whether a minimisation free to choose its supports would also break the symmetry cannot be settled while boundaries are confined to whole cells. It is, however, the direction a real hydrogen chain takes, being unstable to dimerisation [18].

The control on all of it is exact: on a wrapped chain a displacement of one full spacing is a relabelling and must return the same energy, and two independently relaxed runs agree to nine digits. The barrier is also far less sensitive to the one arbitrary length in the construction than the total energy is. Measured on an eight-site chain at a finer mesh, where the pair could be afforded, moving the nuclear cutoff from 0.6 to 0.8  $a_0$  shifts the total by 6.2 eV and the difference by 40 meV, a suppression by a factor of 158. That is the error cancellation of §3 shown rather than argued. It is also the honest size of the uncertainty — some 10% of a barrier of this size, from the cutoff

alone — and it is quoted for a shorter chain than the one scanned above, which is stated because the two are different systems.

**The polarisability is extensive.** With the partition frozen and a uniform field applied, the induced dipole gives  $\alpha$ , and repeating at three chain lengths gives its scaling:

$$\alpha = 7.6 N a_0^3, \quad \text{flat to 2\% across } N = 4, 6, 8.$$

An isolated hydrogen atom has  $\alpha = 4.5 a_0^3$ , so the per-atom value is a modest chain enhancement of it. The scaling is what matters. A response strictly proportional to  $N$  is the signature of charge that cannot leave its site: each domain carries its unit by constraint, so the only responses are the density deforming within a fixed domain and the boundary displacing, both per-atom constants. Charge free to redistribute across the sample would give a response growing faster than  $N$ , and does not.

## 5 A cable of ions

The chain puts the carrier where the attractor is. Conducting matter does not: core electrons occupy the region around each nucleus and the conduction charge stands outside them. Representing that needs two populations, and it is the one arrangement giving a carrier a standoff without violating the tiling, because the inner region is owned rather than left empty.

Sixteen sites at  $a = 4.5 a_0$ , kernels of charge  $+3$ , a closed core of two electrons at each, one valence electron per site outside them: neutral, and with no pseudopotential anywhere — the core is represented by its electrons rather than by a potential standing in for them, so the carrier is kept out of the core region by non-overlap and not by an imposed condition. The core pair is a single homogenised domain of charge two rather than two unit domains meeting at a plane — a closed shell is spherically symmetric, and split halves relax unequally, which the eigenvalue control catches. Its self-repulsion is reduced to match: for a homogenised domain of charge  $q$  the fraction  $1/q$  of its own potential is removed, leaving  $(q-1)/q$  of the naive self-energy, exactly the  $\binom{q}{2}$  pairs the  $q$  electrons form. At  $q = 2$  that is one pair; at  $q = 1$  the self-potential goes entirely, recovering the usual absence of self-interaction. The carriers are  $q = 1$  throughout.

The barrier, evaluated by (3) with windows of  $W = 4$  and  $W = 8$  carriers, is

$$\varepsilon_{\text{bulk}} \approx 246 \text{ meV per carrier,}$$

a factor of 2.6 below the chain. The interaction is the same and the energy functional is the same; what has changed is where the carrier sits relative to its kernel.

**The standoff is set, not found.** The radius  $c_r$  separating cores from carriers is prescribed. It belongs to the specification of the arrangement, alongside the spacing, the kernel charge and the number of electrons per site, and the conduction question is put to the arrangement so specified. Whether a minimisation free to move that interface would settle at the same radius is not decided here: released, the calculation does not converge at whole-cell resolution, and the boundary that fails to settle is the one between neighbouring cores rather than the core-carrier radius itself. The cable is therefore a stated geometry whose conduction is computed, not a configuration the functional produces of itself.

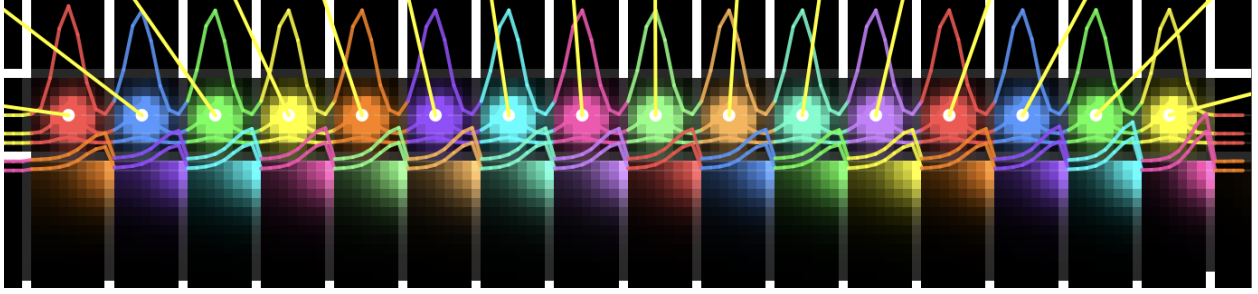


Figure 1: The cable, relaxed. Sixteen sites along the axis, each carrying a kernel of charge  $+3$ , a core domain of two electrons confined within  $c_r = 2.4 a_0$ , and one valence electron outside it. Colour distinguishes domains: the bright lobes on the axis are the cores, the fainter blocks beneath them the valence charge standing off. White verticals mark the free boundaries between neighbouring carriers, and the coloured curves are density profiles along a line through the axis; the yellow segments are net forces on the kernels. The partition shown is the imposed one described in §3 and is held fixed while the wave functions relax, so the boundaries are inputs here rather than unknowns. Displacing the middle  $W$  of these carriers by half a spacing, at fixed cores and kernels, is the quantity computed in §3.

## 6 What sets the pinning

The barrier is not set by the strength of the ion lattice but by whether the carriers bunch into its wells. A carrier distribution that resisted being modulated would ride over the lattice; one that does not resist locks to it.

RealQM’s free interface resists very little, and this is internal to the construction. A flat density is admissible on a free boundary at no cost — a constant  $\psi_i$  satisfies the normalisation, has  $\nabla\psi_i = 0$ , and minimises the electrostatic energy against a neutralising background — so redrawing an interface between domains of equal density costs nothing. The partition therefore offers no resistance to uniform compression, and the only penalty on a *modulated* density is the gradient term in (1), carrying the coefficient  $\frac{1}{2}$ .

**That coefficient is a standardisation, not a constant of the theory.** It is set by calibrating the framework against atomic spectra, hydrogen first among them — on a bound electron in a single potential well. An extended arrangement is a different object, as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering, and the coefficient governing how a carrier lattice resists modulation by an ion lattice has no obligation to equal the one an atom’s spectrum sets.

So the natural thing to ask is how much the answer turns on it, and that is all the following is: a sensitivity, not a calibration. No value was sought that would reproduce a conductivity, a barrier or any other quantity taken from experiment, and none of the numbers below was known in advance. The coefficient is varied from its physical  $\frac{1}{2}$  upward, with the ions, the standoff, the geometry and the mesh unchanged:

|                                   |       |       |      |      |      |
|-----------------------------------|-------|-------|------|------|------|
| coefficient $c$                   | 0.50  | 0.60  | 0.75 | 0.90 | 1.00 |
| $\varepsilon_{\text{bulk}}$ (meV) | 246.1 | 158.7 | 66.4 | 19.9 | 1.0  |
| in units of $kT$                  | 9.5   | 6.1   | 2.6  | 0.77 | 0.04 |

The preferred registry is the same at every point: the carriers sit on their own ion sites throughout, so what falls is the pinning itself and not merely its location. The barrier passes  $kT$  between  $c = 0.75$  and  $c = 0.90$ . A moderate stiffening of the carriers, well short of the doubling, therefore brings this cable out of the range where charge is held and into the range where thermal energy moves it.

The last entry should not be read as a value. One meV is a difference of  $1.5 \times 10^{-4}$  Ha between two total energies of 158 Ha, below what this protocol has been shown to resolve. And the entries away from  $c = \frac{1}{2}$  rest on the two ends  $d = 0$  and  $d = \frac{1}{2}$ , so that the extrema lie there remains an assumption for them — an assumption that failed on the chain of §4, whose minimum is not at registry. It holds for the first entry, which has been scanned: that landscape is monotone from registry to half-spacing, so its endpoints are its extrema and its 246 meV is an amplitude. Whether stiffening moves the extrema has not been tested.

| arrangement                            | barrier per carrier | preferred registry                   |
|----------------------------------------|---------------------|--------------------------------------|
| chain of H atoms                       | 394 meV             | off registry, near $d = \frac{1}{4}$ |
| cable of Li ions, carrier standing off | 246 meV             | carrier on its site                  |
| the same cable, carriers stiffened     | $\approx 1$ meV     | carrier on its site                  |

**A transport number the landscape settles by itself.** A uniform field  $F$  adds  $-NeFd$  to the energy: it tilts the landscape and changes nothing else. The field at which the barrier disappears is therefore a property of  $E(d)$  alone. No clock, no temperature and no dissipation enter it, which is why it can be stated here, among the computed quantities, rather than in §7 with the conductivities. For a sinusoid of amplitude  $V_0$  the steepest slope is  $\pi V_0/a$ , so  $F_{\text{th}} = \pi V_0/ea$ :

| arrangement                 | $V_0$   | $F_{\text{th}}$ ( $\text{V m}^{-1}$ ) |
|-----------------------------|---------|---------------------------------------|
| chain of H atoms            | 394 meV | $5.2 \times 10^9$                     |
| cable of Li ions            | 246 meV | $3.2 \times 10^9$                     |
| the same cable at $c = 0.9$ | 20 meV  | $2.6 \times 10^8$                     |

Above that field charge advances without thermal help; below it, it does not advance at all without. The three divide, and the division is the point. The first two are an order above the dielectric strength of silica,  $\sim 10^9 \text{ V m}^{-1}$ , so those arrangements would break down before they would slide — which is what an insulator does. The stiffened cable is a factor of four below it, so that one would slide first. The same separation that §6 states as a barrier against  $kT$  appears here as a field against a breakdown strength, and neither statement needs a rate theory to make it. The sinusoid enters here as it does in  $K$ , and a scan of  $E(d)$  at small  $d$  would remove it, the threshold then following from the computed slope directly. What does *not* enter is any rate theory.

It is worth saying why the calculation stops at the threshold rather than continuing past it, since driving the system further is the obvious thing to try. Below the threshold a local minimum survives, so the pattern displaces a little and stops: that is an elastic response, the polarisability of §4, and no charge is transferred. Above it there is no minimum left and nothing to stop the slide, so the field does work indefinitely and, in a conservative functional, that energy has nowhere to go. There is no steady state to compute, and therefore no drift velocity. The trap is that the solver will answer anyway. Driven past the threshold the partition does run and walls do advance, so a speed can be read off — but the relaxation is gradient descent in imaginary time and its rate is the descent parameter, chosen rather than derived. The threshold is a property of  $E(d)$  and survives refinement; a current past it is a property of the scheme and does not. Nothing in this article is computed above the threshold.

## 7 What this gives, and what it does not

In the activated regime the corrugation is the activation energy, and the conductivity of a hopping conductor is

$$\sigma = \frac{ne^2a^2\nu}{kT} e^{-E_a/kT}, \quad \nu = \frac{1}{2\pi} \sqrt{\frac{K}{M}}, \quad (4)$$

with  $\sigma$  the conductivity,  $n$  the carrier density,  $e$  the elementary charge,  $a$  the hop distance,  $E_a$  the activation energy,  $k$  Boltzmann's constant,  $T$  the temperature,  $\nu$  the attempt frequency,  $K$  the curvature of the well the carrier sits in and  $M$  the mass of whatever oscillates in it.

**Activated rate theory.** The picture behind (4) is Arrhenius's, made quantitative [14]. A system sits in a potential well separated from the next by a barrier  $E_a$ . It does not cross steadily. It oscillates in the well at the small-oscillation frequency set by the curvature of the well and the mass in it, and on each oscillation there is some chance that thermal fluctuation has put enough energy into the right coordinate for it to get over. The rate is therefore a product of how often it tries and how often a try succeeds:  $\nu$  counts the attempts, and  $e^{-E_a/kT}$  is the fraction of the time the Boltzmann distribution holds at least  $E_a$  there.

The theory carries three conditions, and two of them will matter below. The barrier must be large against  $kT$ : if it is not, crossings are not rare, the system spends a substantial part of its time above the top, and the rate is set by how fast it moves rather than by how seldom it is lucky — so the factorisation into attempts and successes stops describing anything. The system must lose the memory of one attempt before making the next, which in a solid it does by exchanging energy with the lattice. And the barrier must be crossed over rather than through:  $e^{-E_a/kT}$  is a classical occupancy and says nothing about tunnelling.

That last one deserves an answer rather than an assumption, since tunnelling is often what carries charge and the carriers here are electrons. The answer turns on which coordinate crosses. It is not an electron moving through a barrier in space; it is the carrier pattern moving against the ion lattice, and §6 argues that the mass belonging to that motion is ionic, since it is the sites that move. A coordinate that heavy tunnels feebly. Quantitatively, the crossover below which tunnelling overtakes activation is  $T_c = \hbar\omega_b/2\pi k$ , which for the frequencies found here is 179 K for the chain and 43 K for the cable. Both lie below room temperature, so the classical factor is the right one at 300 K — though the chain is close enough to its crossover that quantum corrections would not be negligible in a cold measurement. The argument is tied to the rigid slide: a kink is a lighter object, and whether it tunnels is a question this article does not reach.

What oscillates here is the collective coordinate  $u$ : the displacement of the carrier pattern along the chain, the same variable  $E(u)$  is evaluated against in §3. The attempt is therefore the charge pattern rocking back and forth in one corrugation well against the ion lattice, and since it is the sites that move, that rocking is a lattice vibration. Both  $V_0$  and  $M$  are then taken per carrier, which is consistent: displacing  $m$  carriers rigidly costs  $m$  times the barrier and moves  $m$  times the mass, and  $\nu$  is unchanged.

RealQM supplies  $E_a$ , which decides the temperature dependence and the ratio between arrangements. The prefactor needs  $K$ ,  $M$  and  $n$ , and it is worth being exact about where each comes from.

$K$  is the curvature of the landscape at its minimum, and it is the one place where the shape of that landscape matters rather than its amplitude. Taking the periodic curve to be  $E(u) = \frac{1}{2}V_0[1 - \cos(2\pi u/a)]$  gives  $K = 2\pi^2 V_0/a^2$ , which for  $V_0 = 246$  meV at  $a = 4.5 a_0$  is  $0.80$  eV Å<sup>-2</sup>, and that is the value used below.

The scan of §3 shows the assumption to be wrong, and in a direction worth stating. Across the half period the cable gives, per carrier and relative to registry,

| $d$            | 0 | 0.125 | 0.25 | 0.375 | 0.5  |
|----------------|---|-------|------|-------|------|
| measured (meV) | 0 | -5    | +82  | +184  | +246 |
| sinusoid (meV) | 0 | +36   | +123 | +210  | +246 |

The landscape is monotone from registry to half-spacing, so the amplitude is the difference of its endpoints after all and  $V_0 = 246$  meV stands as a measured quantity rather than an assumed one. But the curve lies below the sinusoid at every interior point, most of all near the minimum, where it is flat out to  $d = 0.125$  instead of rising by 36 meV. The well is therefore much shallower in curvature than a sinusoid of the same amplitude, and since  $\nu \propto \sqrt{K}$  the attempt frequency below is an *overestimate* and the conductivities with it. The steepest slope runs the other way, 6% above the sinusoidal value, so the threshold field is slightly *understated*.

No corrected  $K$  is quoted, because the interior points converge an order of magnitude less well than the endpoints — residuals of 0.04 to 0.08 against 0.004 — and a second difference over a spacing of 0.125 amplifies exactly that. What the scan establishes is the amplitude, and a direction for the prefactor: the flatter the true minimum, the lower  $\nu$  and the lower  $\sigma$ .

There is a second assumption in the same expression, of a different kind, and it should be stated because no scan can remove it. Writing  $\nu = \frac{1}{2\pi} \sqrt{K/M}$  is transition-state theory, and transition-state theory describes one damping regime out of three. Kramers' result is that the prefactor depends on the coupling  $\gamma$  between the moving coordinate and the lattice, and non-monotonically: strongly damped it becomes  $\omega_0 \omega_b / 2\pi \gamma$ , the rate being set by how slowly the coordinate diffuses over the top; very weakly damped it is limited instead by how slowly energy arrives. Only at intermediate damping does  $\gamma$  cancel and the well curvature alone set the rate. That intermediate case is the one assumed here, and it is assumed for a reason worth admitting plainly: it is the only regime whose prefactor can be had from a landscape and a set of masses. The other two require the coupling, which is the same quantity the driven regime needs and which a functional without a boundary velocity cannot supply. One consequence runs against the grain of the rest of this section — transition-state theory *overestimates* a rate, so this prefactor biases  $\sigma$  upward where the upper-bound barrier biases it down. The exponent wins by orders of magnitude and the conductivities remain lower bounds, but the two errors do not point the same way.

$M$  is part of the specification of the system rather than a parameter of the theory. Two masses must be kept apart here. The *electron* mass is genuinely absent from (1): the coefficient of  $|\nabla \psi_i|^2$  is a geometric scale, not  $\hbar^2/2m$ , which is why §6 can vary it and why  $E_a$  comes out without any mass being supplied. The *nuclear* mass is a different kind of thing — a property of the nuclei one has chosen to put in the box. To say the material is lithium is already to say  $Z = 3$  and  $M = 7u$ , exactly as fixing the nuclear positions says what the lattice is. It is needed only when the question turns from what the energy is to how fast something oscillates in it. What moves here is the site, since the charge does not pass between sites but the pattern of domains advances through the lattice, so  $M$  is a nuclear mass and  $\nu$  comes out a lattice frequency.

$n = 1/a^3$  is a convention for reporting rather than a quantity of either kind. It takes the cable's spacing to hold in three dimensions, one carrier per site. The cable is a single line of sites, so this is a stated convention for converting a per-carrier barrier into a bulk conductivity, not something the calculation contains.

With the nuclear mass of the moving species —  $1u$  for the hydrogen chain,  $7u$  for the Li cable — and  $T = 300$  K:

|                  | $E_a$           | $\nu$ (Hz)           | $\sigma$ (S/m)       | for scale       |
|------------------|-----------------|----------------------|----------------------|-----------------|
| chain of H atoms | 394 meV         | $1.8 \times 10^{13}$ | $1.1 \times 10^{-1}$ | intrinsic Ge, 2 |
| cable of Li ions | 246 meV         | $5.5 \times 10^{12}$ | 10                   | intrinsic Ge, 2 |
| cable, stiffened | $\approx 1$ meV | —                    | —                    | — see below     |

The attempt frequencies are  $10^{12}$ – $10^{13}$  Hz, which is where lattice vibrations live: as an energy  $h\nu$ , 23 meV for the cable and 97 meV for the hydrogen chain, the latter high because a proton is light. Nothing was arranged to make that so, and it is the check that the mass entering  $\nu$  is the right kind of mass. The first two rows bracket the conductivity of ordinary semiconducting material, six orders of magnitude apart on a barrier ratio of 2.6.

The third row carries no numbers because it violates the first condition above completely. At  $E_a \approx 1$  meV against  $kT = 25.9$  meV the barrier is  $0.04 kT$  and  $e^{-E_a/kT} = 0.96$ : the carriers are over the top essentially all of the time, so escape is not rare,  $\nu$  counts nothing, and there is no sense in which (4) describes the state. Evaluating it there would return a large conductivity, but the number would be an artifact of applying a rate law where no rate is defined. What that state is, and what law governs it, this calculation does not determine.

These are order-of-magnitude statements and cannot be more. The  $\pm 30$  meV on  $E_a$  is already a factor 3.2 either way in  $\sigma$  at room temperature, the sinusoid is an assumption, and  $M$  and  $n$  are inputs. What the framework delivers is the exponent; that the resulting numbers land in the three named regimes, in the right order, with nothing fitted, is the result.

**What a conductivity requires that a barrier does not.** The division of labour here is the ordinary one. An energy landscape is computed from a theory of charge; a rate is then obtained from it by a theory of statistics laid on top. Density functional theory with harmonic transition-state theory does exactly this for diffusion and catalysis, and nobody regards the second step as a defect of the first. What is unusual in this article is not the import but what sits underneath it: a barrier with nothing fitted in it. It is still worth being exact about the distance between the two, because the barrier is computed and the conductivity is not. Three things must be supplied, and they are of three different kinds.

A *clock*. A barrier says how high, a rate says how often, and the difference is an inertia. The kinetic coefficient in (1) is a geometric scale rather than  $\hbar^2/2m$ , so the electronic functional carries no mass and can set no frequency. This one the framework can nevertheless supply, because the nuclei do move classically:  $\nu$  is a period read off a trajectory rather than a curvature assumed, which is what §8 records as now reachable.

A *temperature*. Equation (1) has no  $T$  in it, no bath and nothing stochastic. The entire apparatus of attempts and escapes, and the factor  $e^{-E_a/kT}$  with it, is imported from activated rate theory and laid over a static minimisation. The model can neither confirm it nor refute it, and this is the largest single import in §6.

A *dissipation channel* — but only in the other regime, and the distinction is worth keeping straight. Below threshold the bath that supplies a fluctuation also takes the energy back between hops, which is why the system returns to a minimum each time and why nothing accumulates; dissipation is implicit in the temperature and needs no separate account. Above threshold there is no barrier to be activated over,  $T$  drops out, and dissipation must instead be explicit, or a carrier merely accelerates and there is no steady drift at all. That regime is not treated here. Its ingredients are not absent in principle — the nuclei are classical, so the electron–lattice channel other treatments import is present in the equations and  $\tau$  would be an output rather than an input — but nothing of the kind is computed.

So the regime this article works in needs two imports, not three: a frequency and a temperature. With those, the drift velocity follows from the landscape directly,  $v_d = (ea^2\nu F/kT) e^{-E_a/kT}$  at small field, every other quantity in it being computed — the barrier, the hop distance, and the density of sites.

Of these, only the temperature is an import in the strong sense — a statistics the framework does not contain and cannot check. The other two are provisional rather than structural: the clock because the nuclei are already dynamical, the dissipation because those same nuclei are the channel through which a carrier would lose its energy. So the present arrangement, a computed landscape with a rate theory brought in from outside, is the expected first stage rather than the end of the matter, and what would replace it is stated in §8.

**The kink, and why the table is a lower bound.** Every barrier above is a rigid slide: all carriers displaced together, so the cost is  $N$  times a per-carrier barrier — which the windowed measurement establishes rather than assumes, the linearity in  $W$  having been tested against a third window — and the whole sample must climb at once. Real conductors do not do this, and no macroscopic sample could.

What a cheaper path saves is not the total work. Both routes run between the same two configurations, which on a wrapped chain are equivalent minima, so each is a climb to a saddle and a return; and moving the same charge the same distance costs at least as much summed along a sequential path as along a simultaneous one. What differs is the height of the saddle — the largest energy that must be present at a single instant — because that is what a thermal fluctuation has to supply in one go, and it is therefore what sets the rate. A sample requiring  $N\varepsilon$  at once is quiet for ever at any  $N$  of physical size; one requiring  $\varepsilon$  at a time,  $N$  times, is not. The mechanical analogue is a carpet, which is dragged whole only against the friction of its whole area but is moved a ruck at a time for less force and, if anything, more total work.

That alone makes the table an upper bound, and by an argument that does not depend on knowing what the sample does instead. The barrier is the minimum, over all paths joining one registry to the next, of the greatest energy along the path; the rigid slide is one such path; a minimum over a set is at most its value at any member. So the numbers above bound the activation energy from above whatever the mechanism turns out to be, and the conductivities bound it from below.

What the cheaper path *is* is not settled here. The carriers on their substrate are a Frenkel–Kontorova chain [15], whose standard cheap path is a kink — a localised compression or rarefaction of the carrier lattice, moving by shifting its centre — and §8 sets out what computing one would require. Until it is computed the gap between the rigid slide and the true barrier is not determined, and neither is the factor by which every  $\sigma$  above is an underestimate. That is the principal limitation of the numbers in this section, and it is worth being plain that it is not a small one:  $\sigma$  depends exponentially on the barrier, so the agreement in the table with intrinsic silicon and germanium is contingent on the bound being reasonably tight, which nothing here establishes.

One consequence is worth recording because it follows from the construction rather than from the kink picture. Doping cannot mean here what it usually means: there is no band to add a carrier to and no gap to promote one across. What can be added is a *domain*, and since the domains tile, an added domain takes its space from its neighbours and compresses the carrier lattice locally while a removed one lets them expand. So  $n$ - and  $p$ -type would be compression and rarefaction, the dopant concentration would set the density of these defects, and conduction would proceed by moving the very defects that doping creates — with an  $n/p$  asymmetry that is structural rather than fitted. None of it is computed, and a dopant at a realistic concentration is one atom in  $10^3$ ,

which sixteen sites cannot represent.

What is outside is not a regime but a kind of quantity. A resistive conductivity  $\sigma = ne^2\tau/m$  needs a momentum relaxation time from electron–phonon scattering, and that needs a current as an object, electron dynamics, and irreversibility; a minimisation over real densities has none of the three, at any barrier height. And band conduction specifically is absent in kind rather than by degree: non-overlap admits no extended states, so there is nothing for a band to be made of, and no choice of stiffness produces one. Neither of these tells us what the unpinned state of §5 is. They say only that if it conducts, it does not do so by filling bands, and that this framework will not supply the rate at which it does.

## 8 Further studies

The nuclei reported here lie on a line. The calculation does not: it is three-dimensional throughout, and what is restricted is the lattice, not the electronic problem. A line is the smallest arrangement in which the question can be posed, and it is why  $n = 1/a^3$  had to be a convention rather than a count. The obvious next step is real configurations in three dimensions: the tetrahedral lattice of silicon and germanium, the close-packed lattices of the alkali metals, the layered structures of the oxides. Nothing in (1) is one-dimensional, and the angular splitting of a kernel’s valence sector already reaches the bonding topologies these lattices are built from. A three-dimensional cell would make the carrier density a property of the structure rather than a stated convention, and would let a barrier be compared with the activation energy of a named material instead of with a range.

**A first three-dimensional lattice.** That step has been taken for the simplest case, and it is worth reporting both for what it settles and for what it does not. Hydrogen has no core and no standoff, so unlike the cable there is no lower bound on how close neighbouring chains may sit — the cable needs  $c_{\text{sep}} \gtrsim 1.6a$ , its cores occupying  $2c_r$  and its carriers needing room between them — and hydrogen atoms can therefore be placed on a simple cubic lattice at the same spacing  $a$  in all three directions. Seventy-two of them,  $3 \times 3 \times 8$  at  $a = 4.5a_0$ , one electron per kernel and nothing imposed beyond the lattice itself.

The construction is verified three ways. At  $1 \times 1$  it reproduces the chain of §4 to all nine digits. The nuclear sum over the 72 protons agrees with an independent evaluation of the same lattice to eleven figures. And the translation control holds exactly: at a matched step count  $E(d=1)$  and  $E(d=0)$  agree to every digit reported, which they must, a displacement of one full spacing being a relabelling of the same calculation.

What it settles is the binding. The lattice gives  $-0.524$  Ha per atom against the chain’s  $-0.453$ , and below even the exact free atom’s  $-0.5$ , because a domain in the lattice is a compact cell of volume  $a^3$  rather than a slab running out to the box face. It is also the better-conditioned calculation, which was not expected of nine times the domains: its residual at  $d = 0$  is  $5.3 \times 10^{-4}$ , against  $1.8 \times 10^{-3}$  for the chain and  $4 \times 10^{-3}$  for the cable, the smallest in this article. A partition into compact cells is evidently closer to what the functional would choose than one into elongated slabs, and  $n = 1/a^3$  is exact for this arrangement where §6 must state it as a convention.

What it does not settle is the barrier, and the record is worth giving in full rather than summarising. The two ends do not relax at the same rate here, where the chain’s two ends agree within a factor of two:

| steps | residual at $d = 0$  | residual at $d = \frac{1}{2}$ | $[E(\frac{1}{2}) - E(0)]/N$ |
|-------|----------------------|-------------------------------|-----------------------------|
| 2000  | $5.4 \times 10^{-4}$ | $5.9 \times 10^{-2}$          | 739 meV                     |
| 4000  | $5.3 \times 10^{-4}$ | $1.1 \times 10^{-2}$          | 650 meV                     |
| 8000  | $5.3 \times 10^{-4}$ | $1.7 \times 10^{-3}$          | 678 meV                     |

The last column is not a barrier and is not labelled as one: it is the global difference between two displacements, unwindowed, so it carries the junctions. It is given to show that the quantity does not settle. The unslid end is converged — it moves by 0.007 meV per electron over the last doubling. The slid end is not, moving by 28, although its residual falls steadily by a factor of six or seven each time. And the barrier itself is not monotone, so it cannot be extrapolated:  $E(\frac{1}{2})$  descends and then rises again, where the same ladder on the cable rose throughout. A fixed-partition relaxation that steps an energy down and then up is not something this article can account for, and until it can, the 4000-step point is better read as an excursion than as a stage.

No barrier is quoted for the lattice. Two things prevent it. The slid end is not converged, as the table shows, and the sequence is not monotone so it cannot be extrapolated. More seriously, the lattice slides *globally*: the windowing of §3 is not applied to it, so its two junctions do not cancel and its difference carries the ends, which is the fault that cost the chain its first number. A value obtained that way is not comparable with the chain’s 394 meV, which is windowed, and setting the two against each other would compare an end-contaminated quantity with a clean one. What the lattice establishes is therefore structural — that the construction is verified three ways, that the binding per atom rises from  $-0.453$  to  $-0.524$  Ha on going from a line to a crystal, and that  $n = 1/a^3$  is exact for it — and not a barrier. Supplying one means windowing the lattice as the chain and cable are windowed, and scanning it.

The scale needed for this is not out of reach, and that is the practical consequence of a cost polynomial of small power in  $N$ . Solids have already been computed in this framework for structural and thermal questions — ionic crystals under melting and under shear, molecular solids including ice, solid nitrogen and carbon dioxide, and a lithium metal lattice of a hundred atoms restoring itself from a perturbation under dynamics [20]. What has not been attempted in any of them is transport. It is also what several of the questions below require rather than merely benefit from. A dopant at a realistic concentration is one atom in  $10^3$  or fewer, so doping cannot be represented on sixteen sites at all — the account of  $n$ - and  $p$ -type in §7 needs a cell large enough to hold one dopant and the lattice around it. The same is true of a kink, whose width is set by the ratio of two curvatures and which must fit inside the cell with bulk on either side of it, and of grain boundaries, interfaces and vacancies, which are where the transport of real material is usually decided.

The one quantity still owed can be discharged two ways. A scan of  $E(d)$  at small  $d$  gives the curvature directly and retires the assumed sinusoid. Better, moving the nuclei under their classical equations of motion — which the framework does, and which is the one genuine clock in it, the relaxation iteration being imaginary time — gives the attempt frequency as a period read off a trajectory, making both the curvature and the assumed shape unnecessary.

The kink is the larger prize. Every barrier here is a rigid slide, so every barrier is an upper bound and every conductivity a lower bound by a factor this article does not determine. Measuring the formation energy  $E_f$  and the Peierls–Nabarro barrier  $E_{\text{PN}}$  would replace bounds with values, and would at the same time make the account of doping in §7 testable:  $n$ - and  $p$ -type as compression and rarefaction of the carrier lattice, dopant concentration setting the kink density, and an  $n/p$  asymmetry that is structural rather than fitted.

One defect in the discretisation blocks more than any single calculation. Positions live on the grid, so a displacement smaller than a cell does not exist, and at the mesh used here one cell is an eighth of a spacing. The defect has two halves. One of them has since been repaired, which is

worth reporting because it changes what is owed and what is merely not yet done.

Nuclear coordinates were truncated to whole cells when handed to the solver, and are now carried as continuous ones. Two quantities return with them. The attempt frequency can be obtained as it was meant to be — by releasing the kernels and reading a period off a trajectory, which retires the assumed curvature altogether — since a thermal oscillation of some  $0.3a_0$  at room temperature is no longer rounded away. And the Peierls instability of a conjugated chain, the distortion that makes polyacetylene a semiconductor and gives its solitons something to be a defect in, is about  $0.03a_0$  per carbon or a sixth of a cell: a chain that previously returned identical energies at five values of the distortion now resolves them. Neither quantity is computed in this article. Both are now reachable, which they were not.

The other half stands, and four quantities wait on it — all of them a boundary moving rather than a nucleus. A domain wall still sits in one cell or the next. The polarisability must therefore be computed with the partition frozen, since a released boundary answers a sub-threshold field by moving zero cells or one. A boundary driven at fixed force either sticks or chatters between adjacent cells, so a depinning threshold cannot be read. The Peierls–Nabarro barrier needs a kink profile finer than a cell. And  $\kappa$ , the second Frenkel–Kontorova constant, requires a non-uniform displacement small enough to be harmonic — smaller than the grid can express, so amplitudes below a cell return the undisplaced chain exactly. Carrying the boundary as a volume fraction rather than by whole-cell ownership would release all four, and is the single change the programme of this section most depends on.

That constant is worth naming as an aim in its own right. The chain of carriers on a periodic substrate is a Frenkel–Kontorova system [15], the standard phenomenological account of commensurate lattices, depinning and kink transport, and its two constants are the substrate amplitude and the coupling. RealQM computes the first:  $V_0$  is the corrugation computed here, which in ordinary use is supplied from data. With the second, the same model yields the kink width, the Peierls–Nabarro barrier and the Aubry pinning criterion [16] with nothing further put in. The transport would then be done by a model whose credentials are not in question, from constants this framework computed rather than fitted — which is a sharper claim than a conductivity resting on an assumed curvature. It is worth noting which standard models are *not* available in the same way: a Hubbard or tight-binding description is degenerate here, since non-overlap makes the transfer integral vanish identically, and Frenkel–Kontorova fits because its degrees of freedom are the ones RealQM has.

Two questions are open in a way that could change the account rather than sharpen it. Whether RealQM has a self-consistent standoff — a cable whose core-carrier radius the functional settles at rather than one supplied with the specification — is the first, and §5 records that it cannot be decided at this resolution, the released calculation being the one place where whole-cell interfaces bite hardest. The second is whether the arrangement in which the barrier all but vanishes conducts in any useful sense, or is merely an unpinned lattice that still holds its charge per site; the polarisability scaling is the test, since charge fixed per site gives  $\alpha \propto N$  whatever the barrier.

Finally, and to state the contact with measurement at its proper strength. No computed number here is set against a measured number for the same system, because these arrangements have no experimental counterpart. What can be said is that three quantities of different kinds land in the ranges measurement gives, none of them arranged to: barriers of 246–394 meV where semiconductor activation is measured at 50–500; attempt frequencies at  $h\nu = 23$  and 97 meV, which are lattice-vibration energies, and would have come out two orders higher had the mass been the electron’s — though computed from a sinusoid’s curvature, which §6 shows to be too sharp, so the frequencies themselves are overestimates even as their scale is right; and threshold fields an order above the dielectric strength of silica, so that these arrangements would fail before they

would slide, as insulators do. That is consistency in magnitude across three independent quantities from a calculation with nothing fitted in it. It is not a test, and it should not be read as one.

A test needs a system that has been measured. The sharpest available is not a value but a slope: the activation energy fixes the temperature dependence of  $\sigma$ , and an Arrhenius plot of a real activated conductor measures the same quantity directly. A conjugated chain such as polyacetylene is the natural candidate — a real material, measured activation energies, transport by the localised defects §7 describes, and doping in the sense described above — and §8 records what remains before it can be computed.

## 9 Conclusion

For a chain of hydrogen atoms, the electron occupying its own kernel’s region, RealQM gives a barrier of 394 meV per electron for the electron pattern to advance one site, and a polarisability  $\alpha = 7.6N$ , extensive in the number of electrons.

For a cable of Li ions with the carriers standing off closed cores it gives 246 meV, a factor of 2.6 less. The two cannot be built from the same ion, and the reason is instructive: a carrier can stand off a core only if there is a core, and since the domains tile, one domain per kernel must contain its kernel — so a standoff requires at least two electrons per site where the chain has one. The difference in ion is entailed by the difference in arrangement rather than chosen alongside it. What is the same is the physics: (1) throughout, with nothing fitted in either.

Stiffening the carriers of that cable lowers the barrier steeply and without changing the preferred registry: at a coefficient of 0.9 against the  $\frac{1}{2}$  an atom sets, it is 20 meV, below room temperature’s  $kT$ , and the charge is no longer held. So the barrier is set less by the ion lattice than by how strongly the carrier distribution resists being modulated by it, which is a property of the interface condition rather than of the interaction.

Taking each barrier as an activation energy and closing the prefactor with an attempt frequency built from the same landscape and a nuclear mass puts the first two arrangements at  $10^{-4}$  and  $10^1 \text{ S m}^{-1}$  — intrinsic silicon and intrinsic germanium — with the attempt frequencies coming out at lattice values unasked. Nothing in this is fitted. What the calculation takes beyond (1) is a specification of the system — which nuclei, at what spacing, under what geometric constraint — together with a convention for reporting a bulk density; and one quantity, the curvature, which is a property of the computed landscape that has been assumed rather than computed, since only the endpoints of  $E(d)$  were run. The conductivities are lower bounds, the rigid slide being one path between the two registries and the barrier the minimum over all of them; by how much is not determined here, since the cheaper path has not been computed. What the framework does not deliver in any regime is a resistive conductivity, whose relaxation time needs a current as an object and irreversible dynamics: a minimisation over real densities has neither.

## A Computational details

All calculations use the RealQM solver `molecule.js` driven by `chain_slide.html` (WebGPU, single GPU). Equation (1) is minimised by imaginary-time relaxation of the densities on a uniform Cartesian grid, with the partition specified by a cell-ownership map.

**The code, and running it.** Everything is at

<https://github.com/Claes542/RealMolecule>

and runs *in a web browser* with nothing to install, compile or configure: the solver is JavaScript and WGSOL, the arithmetic is done by WebGPU on whatever GPU the machine has, and a run is started by opening a page. A WebGPU-capable browser is required — Chrome or Edge at present. Every configuration in this article is a query string, so each calculation below is reproduced by following one link. The reference state and the two windows that give the 246 meV of §5:

[https://claes542.github.io/RealMolecule/chain\\_slide.html?n=16&a=4.5&rc=0&rsing=1.5&cps=8&wrap=1&coax=1&kq=3&ncore=2&chom=1&cr=2.4&steps=4000&d=0](https://claes542.github.io/RealMolecule/chain_slide.html?n=16&a=4.5&rc=0&rsing=1.5&cps=8&wrap=1&coax=1&kq=3&ncore=2&chom=1&cr=2.4&steps=4000&d=0)

and the same with  $\&swin=4\&d=0.5$  and  $\&swin=8\&d=0.5$ , which are the pair the barrier is differenced from; the stiffened runs of §6 add  $\&kins=c$  together with a step count scaled in proportion to  $c$ , for the reason given below. Each run stops at the step count in its own query string and then writes its total energy, its Euler–Lagrange residual and its per-domain eigenvalues into `localStorage` under `realqm_resid`, keyed by the full configuration, so the numbers in the tables below are read out rather than transcribed from the screen. The parameters are named in the source at the top of `chain_slide.html`. For every corrugation the partition is *frozen*: the energy of a stated configuration is evaluated, which is legitimate for configurations related by a symmetry of the lattice and is why translation by one spacing is used as the control. Energies are reported in hartree and converted at  $1 \text{ Ha} = 27211 \text{ meV}$ .

**The chain of hydrogen atoms.**  $N = 16$  atoms in a row at  $a = 4.5 a_0$ , nuclear charge  $+1$ , one electron per site, no pseudopotential. Wrapped labelling, so translation by one spacing is an exact relabelling. Mesh 8 cells per spacing, box  $= (N - 1)a + 10 = 77.5 a_0$ , grid  $138^3$ ,  $h = 0.5616 a_0$ ; nuclear cutoff  $r_{\text{sing}} = 0.6 a_0$ . Fixed 4000 steps. Windowed at  $W = 4$  and  $W = 8$ , so the two junctions cancel in the difference and  $\varepsilon_{\text{bulk}} = [E(8) - E(4)]/4$  is the bulk cost per electron, relative to registry:

| $d$   | $E(W=4)$ (Ha) | $E(W=8)$ (Ha) | $\varepsilon_{\text{bulk}}$ (meV) |
|-------|---------------|---------------|-----------------------------------|
| 0.125 | -5.358520213  | -5.372141768  | -92.7                             |
| 0.250 | -5.372645480  | -5.401376734  | -195.5                            |
| 0.375 | -5.353253672  | -5.347920417  | +36.3                             |
| 0.500 | -5.273676889  | -5.244451806  | +198.8                            |

The displacements available are multiples of  $a/8$ : domain boundaries are whole-cell, so half a cell is not representable and such a run does not converge. Amplitude 394 meV per electron, minimum near  $d = \frac{1}{4}$ , maxima of 0 and +199 meV.

The cutoff dependence was measured separately, on an eight-site chain at the finer mesh of 16 cells per spacing:  $E(0) = -3.623964481$  and  $E(\frac{1}{2}) = -3.435219214$  at  $r_{\text{sing}} = 0.6 a_0$  against  $E(0) = -3.394494680$  and  $E(\frac{1}{2}) = -3.217398798$  at  $0.8 a_0$ , so the totals move by 0.2295 Ha  $= 6.24 \text{ eV}$  while their difference moves by 39.6 meV, a suppression of 158.

Polarisability from the induced dipole at  $F = 5 \times 10^{-4} \text{ au}$  with the partition frozen, at  $N = 4, 6, 8$ :  $\alpha/N = 7.55, 7.63, 7.70 a_0^3$ . The dipole is taken as the difference between the field-on and field-off runs, which removes a small common offset.

**The cable of Li ions.**  $N = 16$  sites at  $a = 4.5 a_0$  on the axis, kernel charge  $+3$ , one core domain per site carrying two electrons and confined within  $c_r = 2.4 a_0$  of the axis, one valence electron per site outside that radius: 32 domains, neutral, and no pseudopotential, so the carrier is excluded from the core region by non-overlap alone. Mesh 8 cells per spacing, box  $= (N - 1)a + 10 = 77.5 a_0$ , grid  $138^3$ ,  $h = 77.5/138 = 0.5616 a_0$ ; the box is the chain span plus a fixed pad and so does not depend on the mesh.

**The nuclear cutoff, and a correction.** A bare  $-Z/r$  well cannot be represented on a grid, so it is clamped at a radius  $R_{\text{sing}}$ . The solver sets  $R_{\text{sing}} = \max(2h, r_{\text{sing}})$  with  $r_{\text{sing}}$  a requested absolute value, and the requested  $0.5 a_0$  used here is smaller than  $2h$  at every mesh employed — 1.12, 0.90 and  $0.75 a_0$  at 8, 10 and 12 cells per spacing. The cutoff was therefore  $2h$  throughout, tied to the grid rather than absolute, and refining the mesh sharpens the ion instead of merely resolving it. An earlier version of this appendix stated that the clamp fixed the potential as the same object at every mesh; that is not what the code does, and any comparison across meshes made under that assumption compares different ions rather than one ion at different resolutions. To hold the potential fixed,  $r_{\text{sing}}$  must be set above  $2h$  at the finest mesh in use.

What the cutoff costs, and where it cancels, is worth recording since it is the largest numerical error here. On a hydrogen atom at  $r_{\text{sing}} = 0.6$  it is 89 mHa, or 2.4 eV — more than any barrier in this article. It cancels between two configurations of one system at one cutoff, which is what a corrugation is: changing it from 0.6 to  $0.8 a_0$  moves the chain’s total by 6.2 eV and its barrier by 40 meV. It does *not* cancel between different systems. Over a cutoff interval of  $0.26 a_0$  the chain moves 123 meV per atom against the free atom’s 46, the chain electron being pressed toward its nucleus by its neighbours and so carrying more density inside the cutoff. Nor can the cutoff simply be shrunk: it is bounded below by  $2h$ , so reducing it means refining the grid, and at the smallest value reachable here the chain comes out more bound per atom than the exact  $\text{H}_2$  molecule, the cutoff by then spanning some two cells. Every energy difference quoted in this article is therefore taken at one cutoff between configurations of one system; no binding energy or comparison with another method’s number is quoted, and none could be without treating the nuclear cusp analytically rather than truncating it. Only the shell slides. All runs use the fixed-step protocol below, at  $r_{\text{sing}} = 1.5 a_0$  (above  $2h = 1.123$ ). Windowed energies at  $d = \frac{1}{2}$ ,  $W$  the number of displaced carriers:

|                                 | $W = 4$     | $W = 8$     |
|---------------------------------|-------------|-------------|
| $E$ (Ha), 4000 steps            | −149.542323 | −149.506148 |
| $E$ (Ha), stiffened, 8000 steps | −157.965427 | −157.965275 |

Equation (3) gives  $\varepsilon_{\text{bulk}} = +246.1$  meV per carrier unstiffened and  $+1.04$  meV stiffened.

**Stiffened carriers.** Identical geometry, mesh and ion lattice, with the coefficient of  $|\nabla\psi_i|^2$  raised from  $\frac{1}{2}$  to 1 for the shell domains only, the cores keeping  $\frac{1}{2}$ . The coefficient enters three places — the imaginary-time step, the  $H\psi$  diagnostic and the energy sum — and all three must carry it or the reported energy does not belong to the state produced. The explicit step requires the diffusion number below  $1/6$ , so the timestep is halved in proportion, and the stiffened runs are given 8000 steps rather than 4000 to cover the same relaxation.

**The fixed-step protocol.** Runs are stopped after a stated number of relaxation steps rather than when the energy stops changing. The energy test — three consecutive frames within  $2 \times 10^{-5}$  Ha — halts at a different step count for every configuration, so two windowed energies would be relaxed to different degrees and their difference, which is the quantity wanted, would mix the two; and a configuration that never satisfies it runs to the step cap, which is weeks. A fixed count makes each run reproducible from its URL, the solver being deterministic to nine digits, and makes the windowed difference a comparison of like with like. Calibration on the cable at  $c_r = 2.4$ ,  $W = 4$ :  $E = -155.354541$  at 200 steps,  $-150.281715$  at 1000,  $-149.615172$  at 2000,  $-149.542323$  at 4000,  $-149.542140$  at 8000. Four thousand steps is used throughout; eight thousand moves the energy by 5.0 meV. The Euler–Lagrange residual plateaus at  $4.0 \times 10^{-3}$  from 2000 steps onward — 0.0233,

0.0059, 0.00404, 0.00403, 0.00403 along the same ladder — while at 2000 steps the energy is still 1.98 eV from its 4000-step value, so the residual detects gross non-convergence but cannot be the sole test.

This ladder also sets the scale against which the stiffened result of §6 must be read. A single total energy still moves 5 meV between 4000 and 8000 steps, and the stiffened barrier is 1 meV. The windowed difference removes what the two configurations have in common, and both are run to the same step count so most of that drift is common; but no test here establishes that the remainder is below 1 meV, which is why §6 claims the collapse and not the value.

**The rate estimate of §6.** Equation (4) evaluated at  $T = 300$  K ( $kT = 25.9$  meV) with  $a = 4.5 a_0 = 2.381$  Å,  $n = 1/a^3 = 7.4 \times 10^{28}$  m<sup>-3</sup>, the nuclear mass of the moving species, and  $K = 2\pi^2 V_0/a^2$  from the assumed sinusoid, giving  $\nu = (1/a)\sqrt{V_0/2M}$ :

| $V_0$ (meV) | $K$ (eV Å <sup>-2</sup> ) | $\nu$ (Hz)           | $\sigma_0$ (S/m)  | $e^{-V_0/kT}$         | $\sigma$ (S/m)        |
|-------------|---------------------------|----------------------|-------------------|-----------------------|-----------------------|
| 1000        | 3.48                      | $5.5 \times 10^{12}$ | $1.4 \times 10^5$ | $1.6 \times 10^{-17}$ | $2.3 \times 10^{-12}$ |
| 240         | 0.84                      | $2.7 \times 10^{12}$ | $7.0 \times 10^4$ | $9.3 \times 10^{-5}$  | 6.5                   |
| 24          | 0.084                     | $8.5 \times 10^{11}$ | $2.2 \times 10^4$ | 0.395                 | $8.8 \times 10^3$     |

Taking  $M = m_e$  instead of an ionic mass raises  $\nu$  to  $6.1 \times 10^{14}$  Hz and  $\sigma$  to  $1.5 \times 10^3$  S m<sup>-1</sup> on the middle row; that frequency is two orders above any lattice vibration, which is the argument for the ionic choice quite apart from the physical one. The  $\pm 30$  meV uncertainty on the middle row is a factor 3.2 in  $\sigma$ .

**Convergence and resolution.** The states entering each windowed difference are relaxed but not fully converged, carrying Euler–Lagrange residuals of about  $4 \times 10^{-3}$ ; the stiffened runs halve the timestep for stability and are less relaxed than the others. Every number reported is for the stated system at the stated mesh, with the nuclear cutoff held at an absolute radius above  $2h$  so that it belongs to the model rather than to the grid. Whether the same numbers survive a different discretisation, or a different implementation of the same equations, is not established here and is left to further work.

**Controls applied.** Translation by one full period must return the same energy: satisfied to nine digits ( $E(0) = E(N)$  exactly) on the wrapped geometries. Repeat runs of one configuration agree to nine digits, so the solver is deterministic and any discrepancy is attributable to the model or the input. Domains related by a symmetry must have equal eigenvalues: this is what exposed the terminal domains of a finite cable binding an electron-volt harder than the interior, and hence the necessity of (3). Euler–Lagrange residuals are  $4 \times 10^{-3}$  for the chain and  $2\text{--}3 \times 10^{-2}$  for the cable.

## On the use of computational assistance

The solver and the drivers used here were written with AI assistance, and the analysis was carried out the same way. Two statements make that disclosure precise, and the second is what licenses the first.

No quantity in the model is calibrated to any measurement. (1) contains Coulomb interaction and kinetic energy and nothing else; what the calculation takes beyond it is a specification of the system — which nuclei, at what spacing, under what geometric constraint — and a stated convention for reporting a bulk density. Nothing was adjusted to reproduce a barrier, a conductivity or any other number, and none of the values reported was known in advance of computing it.

The implementation is checkable against standards independent of it, and was checked. A hydrogen atom must come out at  $-0.5$  Ha; two separated unit charges must repel as  $1/R$ ; translation of a wrapped chain by one spacing is a relabelling and must return the same energy, which it does to nine digits; repeat runs of one configuration must agree, and do, to the last reported digit. That discipline is not a formality. It found, in the course of this work, that the nuclear cutoff was tied to the mesh rather than fixed, so that refining the grid sharpened the ion instead of resolving it; and that nuclear coordinates are truncated to whole cells before the solver sees them, which is why several quantities named in §8 cannot presently be computed at all. Neither is a fitted parameter, and both changed what the calculation means.

The distinction worth drawing is therefore not between assistance with code and assistance with physics, which would suppose that implementation is innocent — it is not, as those two findings show. It is between content calibrated to the data it is meant to explain, which absorbs disagreement, and content derived from a stated principle, which is exposed to it. This article contains none of the first kind.

## References

- [1] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, New York, 1976.
- [2] W. Kohn and L. J. Sham, Self-consistent equations including exchange and correlation effects, *Phys. Rev.* **140**, A1133 (1965).
- [3] J. P. Perdew and M. Levy, Physical content of the exact Kohn–Sham orbital energies: band gaps and derivative discontinuities, *Phys. Rev. Lett.* **51**, 1884 (1983).
- [4] L. Hedin, New method for calculating the one-particle Green’s function with application to the electron-gas problem, *Phys. Rev.* **139**, A796 (1965).
- [5] J. Bardeen and W. Shockley, Deformation potentials and mobilities in non-polar crystals, *Phys. Rev.* **80**, 72 (1950).
- [6] R. A. Marcus, On the theory of oxidation–reduction reactions involving electron transfer. I, *J. Chem. Phys.* **24**, 966 (1956).
- [7] T. Holstein, Studies of polaron motion: Part II. The “small” polaron, *Ann. Phys.* **8**, 343 (1959).
- [8] N. F. Mott, Conduction in non-crystalline materials III. Localized states in a pseudogap and near extremities of conduction and valence bands, *Philos. Mag.* **19**, 835 (1969).
- [9] J. Hubbard, Electron correlations in narrow energy bands, *Proc. R. Soc. London A* **276**, 238 (1963).
- [10] M. Imada, A. Fujimori and Y. Tokura, Metal–insulator transitions, *Rev. Mod. Phys.* **70**, 1039 (1998).
- [11] J. Kirkpatrick *et al.*, Pushing the frontiers of density functionals by solving the fractional electron problem, *Science* **374**, 1385 (2021).

- [12] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner and G. Csányi, MACE: higher order equivariant message passing neural networks for fast and accurate force fields, *Adv. Neural Inf. Process. Syst.* **35**, 11423 (2022).
- [13] D. Pfau, J. S. Spencer, A. G. D. G. Matthews and W. M. C. Foulkes, Ab initio solution of the many-electron Schrödinger equation with deep neural networks, *Phys. Rev. Research* **2**, 033429 (2020).
- [14] P. Hänggi, P. Talkner and M. Borkovec, Reaction-rate theory: fifty years after Kramers, *Rev. Mod. Phys.* **62**, 251 (1990).
- [15] O. M. Braun and Y. S. Kivshar, *The Frenkel–Kontorova Model: Concepts, Methods, and Applications*, Springer, Berlin, 2004.
- [16] S. Aubry, The twist map, the extended Frenkel–Kontorova model and the devil’s staircase, *Physica D* **7**, 240 (1983).
- [17] M. Motta *et al.* (Simons Collaboration on the Many-Electron Problem), Towards the solution of the many-electron problem in real materials: equation of state of the hydrogen chain with state-of-the-art many-body methods, *Phys. Rev. X* **7**, 031059 (2017).
- [18] M. Motta *et al.* (Simons Collaboration on the Many-Electron Problem), Ground-state properties of the hydrogen chain: dimerization, insulator-to-metal transition, and magnetic phases, *Phys. Rev. X* **10**, 031058 (2020).
- [19] C. Johnson, *Real Quantum Mechanics*, <https://realqm.com>.
- [20] C. Johnson, *RealMolecule*: RealQM solver and browser front-ends, <https://github.com/Claes542/RealMolecule>. Runs in a WebGPU browser with no installation; the gallery of live calculations is at <https://claes542.github.io/RealMolecule/gallery.html>.
- [21] C. Johnson, *Real Quantum Mechanics: a new view of the atom*, book manuscript.
- [22] C. Johnson, *A 3D multiphase continuum computational model for atoms*, manuscript; ionisation energies across the periodic table.
- [23] C. Johnson, *RealNucleus: nucleons as charge densities*, manuscript; nuclear binding by dual Coulomb confinement.

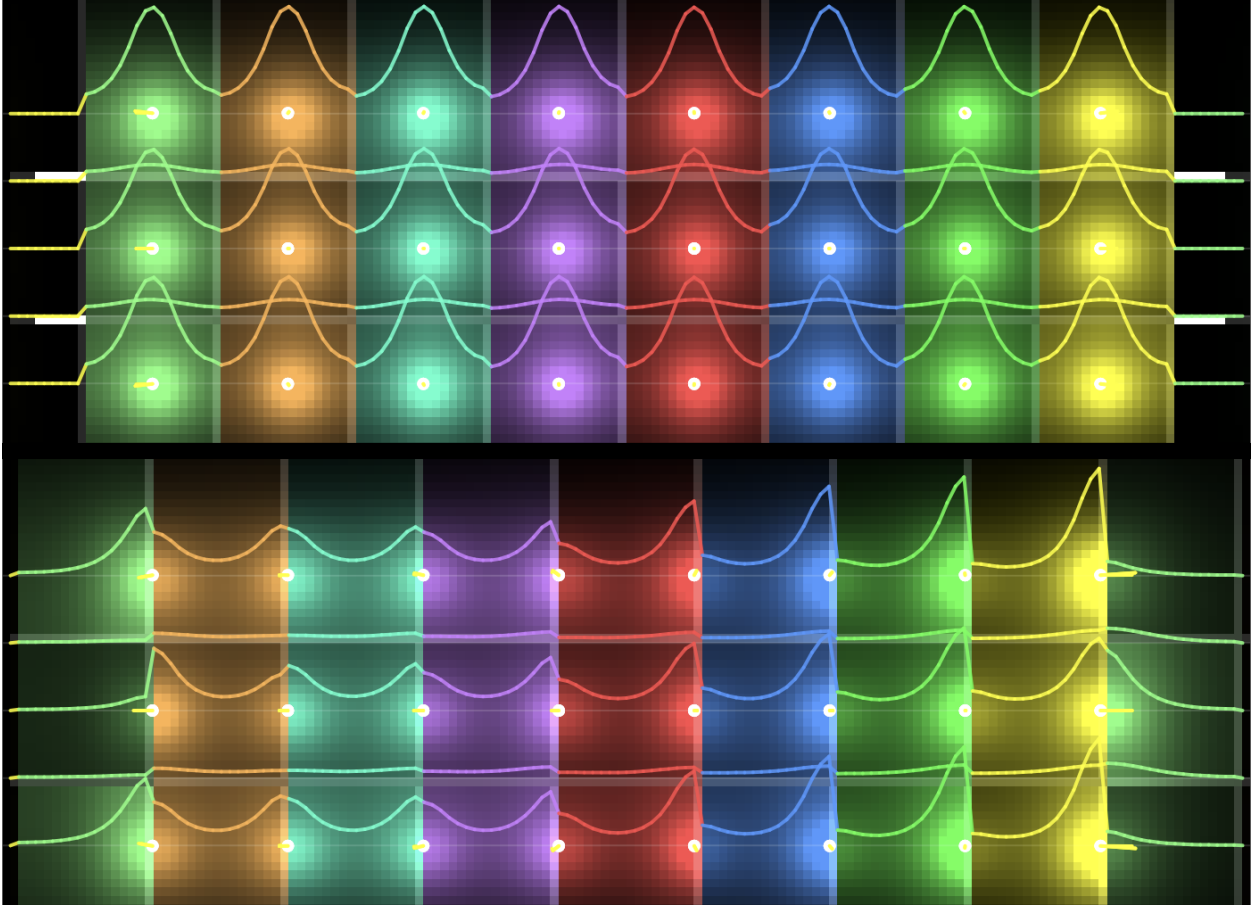


Figure 2: Simple cubic hydrogen,  $3 \times 3 \times 8$  at  $a = 4.5 a_0$ , in its two registries. The frame is a plane through the lattice at fixed  $z$ , so three of the nine chains appear, eight sites each, one electron per kernel and no core, no cutoff and no standoff anywhere in the construction. White dots are the protons, colour distinguishes domains, the white verticals are the boundaries between neighbouring electrons along a chain, and the curves are density profiles along each chain. *Above*,  $d = 0$ : every domain is centred on its own proton and each profile is a single peak on the nucleus. *Below*,  $d = \frac{1}{2}$ : the boundaries now sit on the protons, so each domain spans from one nucleus to the next and its electron is shared between the two bounding it — the profiles go double-humped and the charge piles against the walls. The two differ in nothing but where the boundaries sit, and the energy between them is the quantity this article computes. Both are at the same step count, since two states relaxed to different degrees would not be comparable. Note also that the boundaries run through all three chains at the same  $x$ , which is what makes  $d$  one collective coordinate for the whole lattice rather than one per chain.